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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
not required.
Inclusion and Exclusion Criteria
not required.
Attrition
not required.
Sex as a biological variable
Materials and Methods Antennal Lobe Imaging Antennal lobe glomeruli of male and female P.
Subject Demographics
Age: not required.
Weight: not required.
Randomization
not detected.
Blinding
not detected.
Power Analysis
not detected.
Replication
not required.

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and Algorithms			
Genomes were queried iteratively with TBLASTN, adding newly annotated Polistes ORs to the query file, until no new OR coding regions were identified.	TBLASTN		Suggestion: TBLASTN (RRID:SCR_011822)(link)
To guide annotation of exon-intron boundaries, antennal mRNA from P. fuscatus males and females (gynes) was mapped to P. fuscatus, P. metricus, and P. dorsalis genomes using STAR (Dobin et al. 2013) and assembled into transcripts using Trinity (Haas et al. 2013) (supplementary table S2, Supplementary Material online).	STAR		
	Trinity		Suggestion: Trinity (RRID:SCR_013048)(link)
Predicted transcripts were aligned to genomes using BLAT.	BLAT		Suggestion: BLAT (RRID:SCR_011919)(link)
Uncertain gene models in P. metricus, P. dorsalis, P. canadensis,and P. dominula were aligned to their orthologs in P. fuscatus using Muscle version 3.8.425 with maximum four iterations (Edgar 2004), and gene models were manually adjusted.	Muscle		Suggestion: MUSCLE (RRID:SCR_011812)(link)
All annotation evidence was imported into Geneious v11.1.5 genome browser for manual annotation.	Geneious		Suggestion: Geneious (RRID:SCR_010519)(link)
Transmembrane helices of all putatively functional ORs (>300 amino acids) were predicted using TMHMM version 2.0c (Sonnhammer et al. 1998) and Phobius version 1.01 (Kall et al. 2004).	Phobius		
Phylogenetic Reconstruction Phylogenetic trees were constructed using RAxML (Jones et al. 1992 Stamatakis 2014).	RAxML		
Orthologous genes were determined using OrthoFinder (Emms and Kelly 2015), bootstrap support, and microsynteny.	OrthoFinder		
Values of pairwise d Nd S for orthologs shared by P. fuscatus and P. dorsalis were estimated using PAML version 4.9 (Yang 2007) program yn00 with the Yang and Nielsen (2000) method.	PAML		
Finally, RELAX was run in HyPhy version 2.5.15 to test for relaxed negative (relaxed purifying) selection	HyPhy		

(Wertheim et al. 2015 Pond et al. 2020).			

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For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq